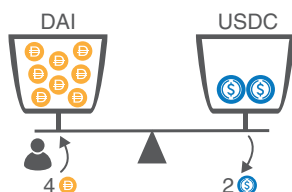


Instantaneous
exchange rate = 1 🟡 = 1 🔵
Invariant (K) = 4 🟡 $\times$ 4 🔵 = 16

Scenario A

Exchange 4 DAI

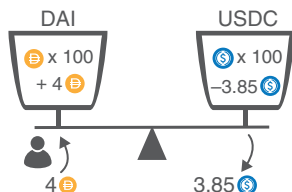


Invariant = $K = 8 \text{ 🟡} \times 2 \text{ 🔵} = 16$
Hence, 4 DAI exchanged for 2 USDC

Scenario B

Exchange 4 DAI

but contract has more liquidity, 100 DAI, 100 USDC



Instantaneous
exchange rate = 1 🟡 = 1 🔵
Before $K = 100 \times 100 = 10,000$
After $K = 104 \times 96.15 = 10,000$
Implied price = 1.04 🟡 = 1 🔵

Figure 6.6 The Mechanics of a Uniswap Automated Market Maker

the market maker). Following the preceding example, higher invariants lead to lower slippage and therefore an increase in effective liquidity. We can think of the invariant as a direct measure of liquidity. In summary, liquidity providing